

4.1 Limits

In many situations, the value of a function at a point is less important than its behavior near that point. For example, consider $f(x) = \frac{x^2-1}{x-1}$, $x \neq 1$. Although $f(1)$ is not defined, we have $f(x) = x + 1$, $x \neq 1$, so the values of $f(x)$ become arbitrarily close to 2 as x approaches 1. Thus, the function exhibits a well-defined behavior near $x = 1$ despite being undefined there.

More generally, when studying continuous and discontinuous functions, we are interested in understanding the behavior of a function as the input approaches a given point from the left or the right. This motivates the concept of the *limit of a function*, which provides a precise mathematical description of the value that a function approaches as its argument approaches a specified point.

Definition 4.1. Let $a \in J \subseteq \mathbb{R}$ be an interval and $f : J \setminus \{a\} \rightarrow \mathbb{R}$ be a function. We say that

$$\lim_{x \rightarrow a} f(x) = \ell$$

if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$x \in J, \quad 0 < |x - a| < \delta \Rightarrow |f(x) - \ell| < \epsilon.$$

In this case, we say that ℓ is the limit of f at a .

Example 4.1. Show that

1. $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = 2$.
2. $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0$.

Solution.

1. Let $\epsilon > 0$ be given. Choose $\delta = \epsilon$. If $0 < |x - 1| < \delta$, then

$$\left| \frac{x^2-1}{x-1} - 2 \right| = |x+1-2| = |x-1| < \delta = \epsilon.$$

$$\text{Hence, } \lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = 2.$$

2. Let $\epsilon > 0$ be given and choose $\delta = \epsilon$. If $0 < |x| < \delta$, then

$$\left| x \sin\left(\frac{1}{x}\right) \right| = |x| \left| \sin\left(\frac{1}{x}\right) \right| \leq |x| < \delta = \epsilon.$$

$$\text{Hence, } \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right) = 0.$$

Theorem 4.2. The limit of a function, if it exists, is unique.

Proof. Suppose $\lim_{x \rightarrow a} f(x) = \ell_1$ and $\lim_{x \rightarrow a} f(x) = \ell_2$. Assume, to the contrary, that $\ell_1 \neq \ell_2$. Choose $\epsilon = \frac{|\ell_1 - \ell_2|}{3} > 0$. Then there exist $\delta_1, \delta_2 > 0$ such that

$$0 < |x - a| < \delta_j \implies |f(x) - \ell_j| < \epsilon, \quad j = 1, 2.$$

Let $\delta = \min\{\delta_1, \delta_2\}$. If $0 < |x - a| < \delta$, then

$$|f(x) - \ell_1| < \epsilon \quad \text{and} \quad |f(x) - \ell_2| < \epsilon.$$

Hence,

$$|\ell_1 - \ell_2| \leq |\ell_1 - f(x)| + |f(x) - \ell_2| < 2\epsilon = \frac{2}{3}|\ell_1 - \ell_2|,$$

which is impossible. Therefore, $\ell_1 = \ell_2$. $\square$

Theorem 4.3 (Algebra of Finite Limits). Let $J \subseteq \mathbb{R}$ be an interval, let $a \in J$, and let $f, g : J \setminus \{a\} \rightarrow \mathbb{R}$ be functions. Assume that

$$\lim_{x \rightarrow a} f(x) = \ell_1 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \ell_2,$$

where $\ell_1, \ell_2 \in \mathbb{R}$. Let $\alpha \in \mathbb{R}$. Then:

1. $\lim_{x \rightarrow a} (f(x) + g(x)) = \ell_1 + \ell_2$.
2. $\lim_{x \rightarrow a} (\alpha f(x)) = \alpha \ell_1$.
3. $\lim_{x \rightarrow a} (f(x)g(x)) = \ell_1 \ell_2$.
4. If $\ell_1 \neq 0$, then there exists $\delta > 0$ such that $f(x) \neq 0$ whenever $0 < |x - a| < \delta$. Moreover,

$$\lim_{x \rightarrow a} \frac{1}{f(x)} = \frac{1}{\ell_1}.$$

Definition 4.2 (Right-Hand Limit). Let $J \subseteq \mathbb{R}$ be an interval and $a \in J$ be not the right endpoint of J . Suppose $f : J \setminus \{a\} \rightarrow \mathbb{R}$ be a function. We say that

$$\lim_{x \rightarrow a^+} f(x) = \ell$$

if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$x \in J, \quad 0 < x - a < \delta \Rightarrow |f(x) - \ell| < \epsilon.$$

In this case, ℓ is called the **right-hand limit** of f at a .

Definition 4.3 (Left-Hand Limit). Let $J \subseteq \mathbb{R}$ be an interval and $a \in J$ be not the left endpoint of J . Suppose $f : J \setminus \{a\} \rightarrow \mathbb{R}$ be a function. We say that

$$\lim_{x \rightarrow a^-} f(x) = \ell$$

if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$x \in J, \quad 0 < a - x < \delta \Rightarrow |f(x) - \ell| < \epsilon.$$

In this case, ℓ is called the **left-hand limit** of f at a .

Example 4.4. Let

$$f(x) = \begin{cases} 0, & \text{if } x > 0, \\ 1, & \text{if } x < 0. \end{cases}$$

Then

$$\lim_{x \rightarrow 0^+} f(x) = 0, \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = 1.$$

Theorem 4.5. Let $J \subseteq \mathbb{R}$ be an interval and $a \in J$ be neither the left endpoint nor the right endpoint of J . Suppose $f : J \setminus \{a\} \rightarrow \mathbb{R}$ be a function. Then

$$\lim_{x \rightarrow a} f(x) = \ell$$

if and only if

$$\lim_{x \rightarrow a^-} f(x) = \ell \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) = \ell.$$

Theorem 4.6 (Sequential Characterization of Limits). Let $a \in J \subseteq \mathbb{R}$ be an interval and $f : J \setminus \{a\} \rightarrow \mathbb{R}$ be a function. Then $\lim_{x \rightarrow a} f(x) = \ell$ if and only if for every sequence (x_n) in $J \setminus \{a\}$ satisfying $x_n \rightarrow a$, we have $f(x_n) \rightarrow \ell$, as $n \rightarrow \infty$.

Example 4.7. Show that

$$\lim_{x \rightarrow 2} (x^2 + 1) = 5.$$

Solution. Let (x_n) be any sequence such that $x_n \rightarrow 2$. Then, by the algebra of convergent sequences,

$$x_n^2 + 1 \rightarrow 2^2 + 1 = 5.$$

Hence, by the Sequential Characterization of Limits,

$$\lim_{x \rightarrow 2} (x^2 + 1) = 5.$$

Exercise 4.3. Let

$$f(x) = \begin{cases} 0, & \text{if } x \in \mathbb{Q}, \\ 1, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Show that

$$\lim_{x \rightarrow a} f(x)$$

exists only for $a = 0$. *Hint.* Use the density of the rational and irrational numbers in $\mathbb{R}$. Alternatively, use the Sequential Characterization of Limits by considering one sequence of rational numbers and one sequence of irrational numbers, both converging to a .

4.2 Infinite Limits

Definition 4.4 (Infinite Limits). Let $a \in J \subseteq \mathbb{R}$ be an interval and $f : J \setminus \{a\} \rightarrow \mathbb{R}$ be a function.

1. We say that

$$\lim_{x \rightarrow a} f(x) = +\infty$$

if for every $M > 0$, there exists $\delta > 0$ such that

$$x \in J, \quad 0 < |x - a| < \delta \Rightarrow f(x) > M.$$

2. We say that

$$\lim_{x \rightarrow a} f(x) = -\infty$$

if for every $M < 0$, there exists $\delta > 0$ such that

$$x \in J, \quad 0 < |x - a| < \delta \Rightarrow f(x) < M.$$

Example 4.8. Show that $\lim_{x \rightarrow 2} \frac{1}{(x-2)^2} = +\infty$.

Solution. Let $M > 0$ be given. Choose $\delta = \frac{1}{\sqrt{M}}$. If $0 < |x - 2| < \delta$, then

$$\frac{1}{(x-2)^2} > M.$$

Hence, by the definition of an infinite limit,

$$\lim_{x \rightarrow 2} \frac{1}{(x-2)^2} = +\infty.$$

Exercise 4.4. Solve the following exercises.

1. Use the sequential criterion for limits to show that the following limits do not exist.

(i) $\lim_{x \rightarrow 0} \cos \frac{\pi}{x}.$

(ii) $\lim_{x \rightarrow 0} \frac{1}{x} \sin \frac{1}{x}.$

2. Let

$$f(x) = \begin{cases} x, & x \in \mathbb{Q}, \\ -2 - x, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Show that

(i) $\lim_{x \rightarrow -1} f(x) = 1.$

(ii) $\lim_{x \rightarrow c} f(x)$

does not exist whenever $c \neq -1$.

3. Show that the following limits do not exist.

(i) $\lim_{x \rightarrow 0} \frac{\sin |x|}{\sin x}.$

(ii) $\lim_{x \rightarrow 0} \frac{2x + |x|}{x^2 - |x|}.$

4. Evaluate the following limits.

(i) $\lim_{x \rightarrow 0^+} \sqrt{x - [x]}, \quad \lim_{x \rightarrow 0^-} \sqrt{x - [x]}.$

(iii) $\lim_{x \rightarrow 0^+} ([x] + [1-x]), \quad \lim_{x \rightarrow 0^-} ([x] + [1-x]).$

(ii) $\lim_{x \rightarrow 0^+} \frac{1}{e^x + \frac{1}{x}}, \quad \lim_{x \rightarrow 0^-} \frac{1}{e^x + \frac{1}{x}}.$

(iv) $\lim_{x \rightarrow 0^+} x \left[\frac{1}{x} \right], \quad \lim_{x \rightarrow 0^-} x \left[\frac{1}{x} \right].$

5. Evaluate the following limits.

(i) $\lim_{x \rightarrow \infty} \frac{\sin x}{e^x \cos x}.$

(ii) $\lim_{x \rightarrow \infty} (x - \sqrt{x^2 - 1}).$

(iii) $\lim_{x \rightarrow \infty} (\sqrt[3]{x+1} - \sqrt[3]{x}).$

5 Continuity

The concept of a limit describes the behavior of a function as the input approaches a point. A natural question is whether the value of the function at that point agrees with the value suggested by its nearby behavior. If this happens at every point in its domain, the function is said to be *continuous*.

Intuitively, a continuous function has no sudden jumps, breaks, or holes in its graph. Small changes in the input produce only small changes in the output. This idea is fundamental in mathematics and its applications, as many physical phenomena, such as the motion of an object, the flow of heat, and population growth, are naturally modeled by continuous functions. The precise definition of continuity is therefore expressed in terms of limits.

Definition 5.1. Let $a \in J \subseteq \mathbb{R}$ and $f : J \rightarrow \mathbb{R}$ be a function. We say that f is **continuous at a** if for every $\epsilon > 0$, there exists $\delta > 0$ such that

$$x \in J \text{ and } |x - a| < \delta \implies |f(x) - f(a)| < \epsilon.$$

We say that f is **continuous on J** if it is continuous at every point of J .

Example 5.1. Show that the following functions are continuous.

1. $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x$, for all $x \in \mathbb{R}$.
2. $f : [-R, R] \rightarrow \mathbb{R}$, $f(x) = x^2$, for all $x \in [-R, R]$.

Solution.

1. Let $a \in \mathbb{R}$ and let $\epsilon > 0$ be given. We have

$$|f(x) - f(a)| = |x - a|.$$

Thus, it is natural to choose

$$\delta = \epsilon.$$

Then, whenever $|x - a| < \delta$, we obtain

$$|f(x) - f(a)| = |x - a| < \delta = \epsilon.$$

Hence, f is continuous at a . Since a was arbitrary, f is continuous on $\mathbb{R}$.

2. Let $a \in [-R, R]$ and let $\epsilon > 0$ be given. Observe that

$$|f(x) - f(a)| = |x^2 - a^2| = |x + a||x - a| \leq (|x| + |a|)|x - a| \leq 2R|x - a|.$$

Choose

$$\delta = \frac{\epsilon}{2R}.$$

Then, whenever $|x - a| < \delta$, we have

$$|f(x) - f(a)| \leq 2R|x - a| < 2R\delta = \epsilon.$$

Hence, f is continuous on $[-R, R]$.

Exercise 5.1. Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$, defined by $f(x) = x^2$, is continuous on $\mathbb{R}$ using the ϵ - δ definition. Hint. Observe that $|x^2 - a^2| = |x - a||x + a|$. To estimate $|x + a|$, first assume $\delta \leq 1$. Then, whenever $|x - a| < \delta$, we have

$$|x| \leq |x - a| + |a| < 1 + |a|,$$

and hence

$$|x + a| \leq |x| + |a| < 1 + 2|a|.$$

Now, choose $\delta = \min \left\{ 1, \frac{\epsilon}{1+2|a|} \right\}$.

5.1 Sequential Criterion for Continuity

Definition 5.2. Let $a \in J \subseteq \mathbb{R}$ and $f : J \rightarrow \mathbb{R}$ be a function. We say that f is **continuous at** a if for every sequence (x_n) in J with $x_n \rightarrow a$, we have $f(x_n) \rightarrow f(a)$. We say that f is **continuous on** J if it is continuous at every point of J .

Example 5.2. 1. Show that every constant function is continuous.

2. Show that the identity function $f : J \rightarrow \mathbb{R}$, defined by $f(x) = x$ for all $x \in J$, is continuous.

3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Then f is not continuous at 0.

Solution.

1. Let $J \subseteq \mathbb{R}$ and $f : J \rightarrow \mathbb{R}$ be the constant function defined by $f(x) = c$ for all $x \in J$.

Fix any $a \in J$, and (x_n) be a sequence in J such that $x_n \rightarrow a$. Since $f(x_n) = c$ for every n , the sequence $(f(x_n))$ is constant. Hence,

$$f(x_n) \rightarrow c = f(a).$$

Therefore, f is continuous at a . As a was arbitrary, we conclude that f is continuous on J .

2. Fix any $a \in J$, and let (x_n) be a sequence in J such that $x_n \rightarrow a$. Since $f(x_n) = x_n$ for every n , we have

$$f(x_n) = x_n \rightarrow a = f(a).$$

Hence, f is continuous at a . Since a was arbitrary, f is continuous on J .

More generally, for every $n \in \mathbb{N}$, the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^n$ is continuous.

3. Consider the sequence $x_n = \frac{1}{n}$. Then $x_n \in \mathbb{Q}$ and $x_n \rightarrow 0$. Hence

$$f(x_n) = 1 \rightarrow 1 = f(0).$$

Now let

$$y_n = \frac{\sqrt{2}}{n}.$$